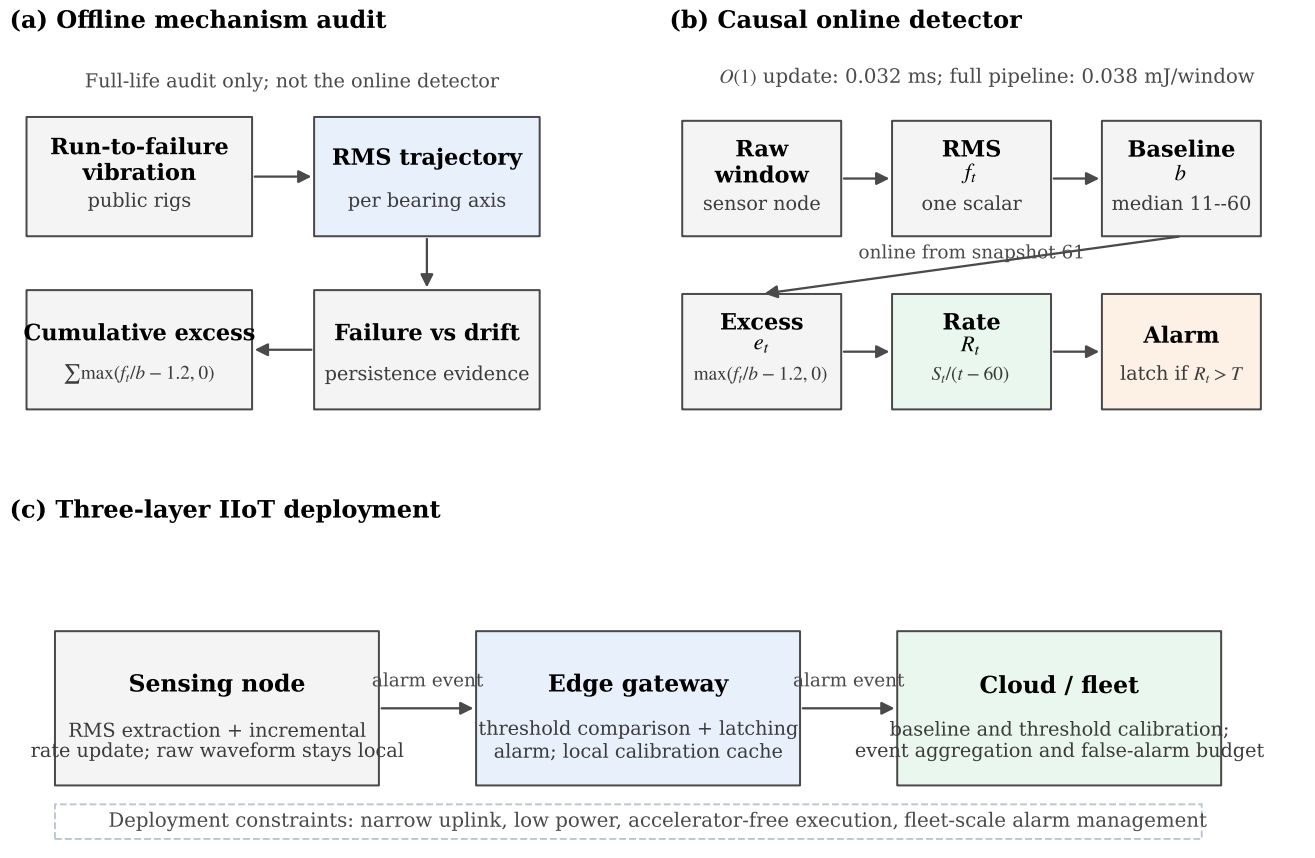
(a) Offline mechanism audit
Full-life audit only; not the online detector
Run-to-failure vibration
public rigs
RMS trajectory
per bearing axis
Failure vs drift
persistence evidence
Cumulative excess
$\sum \max(f_t/b - 1.2, 0)$
(b) Causal online detector
$O(1)$ update: 0.032 ms; full pipeline: 0.038 mJ/window
Raw window
sensor node
RMS
f_t
one scalar
Baseline
b
median 11--60
online from snapshot 61
Excess
e_t
$\max(f_t/b - 1.2, 0)$
Rate
R_t
$S_t/(t-60)$
Alarm
latch if $R_t > T$
(c) Three-layer IIoT deployment
Sensing node
RMS extraction + incremental rate update; raw waveform stays local
alarm event
Edge gateway
threshold comparison + latching alarm; local calibration cache
alarm event
Cloud / fleet
baseline and threshold calibration; event aggregation and false-alarm budget
Deployment constraints: narrow uplink, low power, accelerator-free execution, fleet-scale alarm management